

Using SQL for Regression Analysis

This article is a sequel to our earlier one on using SQL for data analysis. In case they've missed it, readers are advised to read the earlier article published in the February 2020 issue of *OSFY*.



Comparing different parameters of a data set is a common practice to derive a meaningful conclusion. The sample data may be something like rainfall vs agricultural production, stock-index vs GDP or other such data from business and academic fields. Different prediction techniques are used for these studies to estimate their relationships. Nowadays, plenty of ready-to-use computational solutions are available and due to the easy availability of IT infrastructure, even those without tech degrees are using these tools for their own benefit.

To identify the relationship between two or more related features, correlation and regression analyses are commonly used. Regression analysis explores the existence of any linear relationship while correlation establishes the strength of that linear relationship. A linear regression is a statistical

model that analyses the relationship between a response variable (often called *y*) and one or more predictor variables and their interactions (often called *x* or explanatory variables). This technique is effective and easily understandable unlike other data analysis algorithms like artificial neural networks, decision trees or random forests.

Basic descriptive statistics

To have a simple discussion on basic statistics, let us consider a customer table, containing the customer ID, age, income and average spending patterns. Using SQL, it is easy to get basic statistical values like the mean, median and mode of the different parameters of the customer table.

Average: Basic statistical parameters like the mean, median and mode can be computed using SQL functions either with built-in functions or from derived queries. For instance, to get